

Problem 4: The necessity of Non-linear activations

- 3 layer neural network with out activation function

input $\rightarrow$ vector x
 First hidden layer $\rightarrow W^{[1]} b^{[1]}$
 output $\rightarrow W^{[2]} b^{[2]}$

x shape $(1, m)$
 W shape (m, k)
 b shape $(1, k)$

$$a) \quad z^{[1]}_{(1, k)} = x_{(1, m)} W^{[1]}_{(m, k)} + b^{[1]}_{(1, k)}$$

$$b) \quad \hat{y}_{(1, q)} = z^{[1]}_{(1, k)} W^{[2]}_{(k, q)} + b^{[2]}_{(1, q)}$$

$$c) \quad \text{from part a)} \quad \hat{y} = z^{[1]} W^{[2]} + b^{[2]} \quad \xrightarrow{z^{[1]} = x W^{[1]} + b^{[1]}}$$

$$\begin{aligned} \hat{y} &= (x W^{[1]} + b^{[1]}) W^{[2]} + b^{[2]} \\ &= x W^{[1]} W^{[2]} + b^{[1]} W^{[2]} + b^{[2]} \\ &= x_{(1, m)} W^{[1]}_{(m, k)} W^{[2]}_{(k, q)} + b^{[1]}_{(1, k)} W^{[2]}_{(k, q)} + b^{[2]}_{(1, q)} \end{aligned}$$

can be written as $\hat{y} = W' x + b'$

d) Show above expression

$$\begin{aligned} \rightarrow \hat{y} &= (x W^{[1]} + b^{[1]}) W^{[2]} + b^{[2]} \\ \hat{y} &= x W^{[1]} W^{[2]} + b^{[1]} W^{[2]} + b^{[2]} \end{aligned}$$

$W' = W^{[1]} W^{[2]}$
 multiplier
 $(m, q) = (m, k) (k, q)$

$b' = b^{[1]} W^{[2]} + b^{[2]}$
 constant +
 $(1, q) = (1, k) (k, q) + (1, q)$

$$\hat{y} = x W' + b' \quad \text{shapes: } (1, q) = (1, m) (m, q) + (1, q)$$

e) Non linear activation functions are essential to give NN their representational power as without these activation functions, even if we have multiple NN layers like this example, they will be turned to one linear layer and will be equivalent to single linear transformation $\hat{y} = W'x + b$. By adding non-linear activation functions, NN model can learn complex boundaries.

(composition of linear maps is a linear map. A 3-layer network has same representation power of a single linear layer.)